

The argument structure of Turkish verbal reflexives

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Abstract: We leverage the rich diagnostic possibilities available in Turkish to probe the representation of verbal reflexivity in this language. We show that the sole syntactic argument of Turkish verbal reflexives, the surface subject, shows evidence of a VP-internal origin, but also passes diagnostics otherwise only satisfied by deep subjects. We argue that this mixed behavior reflects a movement derivation: a reflexive flavor of the agent-introducing head Voice attracts the sole argument from a low base position to its specifier and derives the reflexive interpretation through VoiceP-internal binding of a free variable. Phrasal movement — a *bona fide* narrow-syntactic operation — can then be crucially implicated in the derivation of verbal reflexivity.

Keywords: verbal reflexives; argument structure; A-movement; Voice; Turkish

1 Introduction

We discuss the syntax of Turkish verbal reflexives (TVRs), (1). The verb in this example supports a passive reading, according to which *Leyla* was washed by a disjoint agent; but also a reflexive reading, whereby *Leyla* is both the agent and the theme of the washing. Past studies of TVRs (Key, 2023, 2024) leave open the provenance of the surface subject of these verbs, our main focus here.

- (1) Leyla soğuk su-yla yıka-**n**-dı-Ø.
Leyla.NOM cold water-with wash-NACT-PST-3SG
‘Leyla washed herself with cold water.’ OR ‘Leyla was washed with cold water.’

The general descriptive profile of these verbs is common cross-linguistically (e.g. Geniušienė, 1987; Kemmer, 1993; Reuland, 2018; Alexiadou and Schäfer, 2013): (1) involves a single overt argument, the surface subject; and a particular kind of voice morphology, also borne by passives in the language, with verbal reflexivity never being possible with morphologically active verbs, (2).

- (2) Leyla soğuk su-yla yıka-dı-Ø.
Leyla.NOM cold water-with wash-PST-3SG
‘Leyla washed (something) with cold water’; NOT ‘L was washed’ or ‘L washed herself’

As in other languages, the reflexivization strategy in (1) is restricted in two ways not typical of pronominal reflexives in the language. TVRs are lexically restricted: not every Root forms a verbal reflexive. They are also thematically restricted in the sense that verbs incompatible with Agents (e.g.

experiencer predicates) cannot form verbal reflexives (for overview see, [Reuland, 2018](#)).

A first question raised by verbal reflexives concerns the extent to which they involve pronominal anaphora of some kind: to what extent is (1) similar to (3), where the source of reflexivity is a pronominal reflexive co-indexed with a distinct nominal? A second question concerns what happens if it turns out that the two strategies are dissimilar: what does it mean to be a verbal reflexive? Both questions require precision on how reflexivity is encoded in the reflexive parse of (1), that is, what guarantees that *Leyla* is ultimately understood both as the agent and as the theme of the washing.

- (3) Leyla **kendin-i** soğuk su-yla yıka-dı-Ø.
Leyla.NOM self-ACC cold water-with wash-PST-3SG
'Leyla washed herself with cold water.'

Different approaches maintain that the phrasal syntax plays a central role in deriving reflexive interpretations not just in (3)—where the centrality of syntax to pronominal reflexivity is uncontroversial (see e.g. [Chomsky, 1973, 1981](#); [Reuland, 2011](#); [Reinhart, 1983](#); [Reinhart and Reuland, 1993](#); [Büring, 2005](#); [Safir, 2004](#))—but also in (1). Many syntactic approaches employ *movement* as an operation crucially implicated in the derivation of reflexivity, as discussed below.

We argue that TVRs instantiate a case where a movement-based derivation of verbal reflexivity is evidenced with particular clarity. Turkish makes available a cluster of phenomena that help diagnose with unusual precision the derivational history of the surface subject of verbs like the one in (1). We first show (in §2) that such verbs are syntactically intransitive: in (1), *Leyla* is the only syntactic argument in the structure. We then show (in §3) that the surface subject originates as a deep object; TVRs thus have the basic configurational syntax of unaccusatives/passives, with the deep object ultimately raising to become the grammatical subject. Interestingly, however, we demonstrate in §4 that this argument transits through an intermediate derived subject position in the specifier of VoiceP; crucially, this intermediate step is only evidenced for reflexives, and not for passives or unaccusatives. We argue that these observations, which arise from diagnostics sensitive to syntactic position and not merely to thematic roles, support movement as a syntactic route to reflexivization in natural language.

At the same time, the facts demand a particular view of movement-based identification. Turkish shows that verbal and pronominal reflexives pattern in crucially different ways, such that the movement-based derivation cannot be extended to pronominal reflexives; there is no anaphor to bind in the syntax of TVRs, so the derivation of the reflexive reading cannot involve a pronominal element *per se*; and it can be shown that the movement step responsible for reflexivization is not to the grammatical subject position, but to a lower VoiceP-internal derived position.

Existing movement-approaches come in three variants, none of which suffices for our data. A

Our analysis is previewed in (4) and (5). We argue that the agent-introducing head Voice comes in a reflexive variant, Voice_{REFL} (Labelle 2008; McGinnis 2022; Paparounas 2023; Kallulli and Roberts 2025; Oikonomou and Alexiadou 2022; Raghotham 2022; cp. Ahn 2015; Kratzer 2009; Baker 2022; Paparounas and Akkuş 2024 for related ideas in *pronominal* reflexivity). Voice_{REFL} does not introduce an externally merged specifier, but it triggers movement of a lower element – typically, the complement of the complex head *v* – to its specifier. (We adopt acategorial Roots head-adjoined to categorizers; alternatives would leave the analysis unchanged). At LF, Voice_{REFL} introduces an Agent role saturated by a free variable; movement of a lower element to the specifier of Voice results in VoiceP-internal binding of this free variable, effectively reflexivizing the Agent role. (Throughout, we notate movement to Spec,TP only when relevant; since the denotations of Asp and T are not crucial, our LF trees are limited to the VoiceP).



The bigger-picture upshot of our discussion concerns the role of syntax: since movement is a *bona fide* narrow-syntactic operation, TVRs must be constructed syntactically. All things being equal, this result is compatible both with analyses of verbal reflexives couched in theories that eschew pre-syntactic word-formation across the board (Kastner, 2017; Alexiadou and Schäfer, 2013; Wood, 2014; Paparounas, 2023; McGinnis, 2022; Spathas et al., 2015); and with theories that countenance syntactic reflexivization as a parametric option alongside a lexical strategy (Reinhart and Siloni 2004, 2005; Siloni 2012; Papangelis 2004; cp. Lidz 1996, 2001a,b; Bresnan 1982; Wehrli 1986; Grimshaw 1982; Bouchard 1982; Chierchia 2004; and see Marelj and Reuland 2016 for a reduction of the parameter). We state our analysis in the vocabulary of the former type of theory; at a minimum, our analysis can be seen as giving substance, for a language with the properties of Turkish, to the notion of *syntactic bundling* invoked in Reinhart and Siloni (2004, 2005).

We do not attempt to adjudicate further between the two classes of theories, reserving broader architectural comparisons for a different occasion. We note, however, that the fact that TVRs demand a syntactic analysis may ultimately prove informative as to the criteria that the latter type of theory uses to classify a given reflexivization strategy as ‘lexical’ or ‘syntactic’ in the first place. For instance, lexical restrictedness is taken to be a hallmark of ‘lexical’ reflexivization in Reinhart and Siloni (2005); but TVRs are built syntactically despite the fact that not every Root in Turkish can result in a TVR. From the perspective of a syntactic theory of word-formation, this is not surprising: nothing in a theory combining Roots with verbal functional structure demands that every combination of the two be licit (see e.g. Marantz, 1997; Alexiadou and Schäfer, 2013; Alexiadou and Doron, 2012; Alexiadou and Anagnostopoulou, 2004; Alexiadou et al., 2006; Schäfer, 2008a; Pylkkänen, 2008; Alexiadou and Doron, 2012; Borer, 2013; Alexiadou et al., 2015; Wood, 2015; Kastner, 2017; Jo, 2019).

Note, finally that the notion of lexical restrictedness itself must be made precise. Like verbal reflexives in other languages, TVRs have been described as being formed only from a set of Roots which is *i)* small in cardinality and *ii)* lexicosemantically circumscribed to predicates denoting different kinds of self-action, such as grooming verbs (see e.g. Kornfilt 1997:140; Kemmer 1993; Geniušienė 1987). Our investigation has revealed that, for Turkish at least, the class of Roots that form verbal reflexives is in fact neither small nor clearly lexicosemantically constrained, even on a generous conception of the notion of self-orientedness. Recent work on the language reaches the same conclusion (see Arslan 2022 for TVRs and Atlamaz and Öztürk 2023 for Turkish verbal reciprocals; other languages are noted to behave similarly, e.g. Dutch in Reuland 2011, 199). There are hundreds of TVR-forming Roots, of which many if not most are not ‘inherently’ reflexive in any obvious way (e.g.

bağla-n ‘attach’, *silkele-n* ‘dust off [figuratively]’, *salla-n* ‘swing (e.g., in a swing)’, *siğ-in* ‘squeeze’, *sak-in* ‘avoid/protect oneself from’).¹ If the set of Roots capable of forming TVRs is not predictable solely on the basis of lexical semantics, then any analysis must enumerate this set of Roots explicitly; on our analysis, selectional restrictions will regulate the combination of reflexive Voice with particular Root+*v* combinations (see section 5).

As for thematic restrictions, viz. the fact that TVRs only derive agent-oriented reflexives, this type of restriction is robust in Turkish. Towards the end of the paper, we discuss how it follows from an account with Reflexive Voice, and how our specific account may generate predictions on the range of roles that can be Voice-identified with the agent role.

2 Intransitivity and monadicity

Numerous diagnostics distinguish transitive verbs with pronominal reflexive arguments from verbal reflexives; we provide a select few indicative tests here. Some are interpretive tests, teasing apart dyadic from monadic predicates (Abusch, 1989; Reuland, 2001; Reinhart and Siloni, 2004, 2005; Reinhart and Reuland, 1993; Jackendoff, 1992; Labelle, 2008; Safir, 2004; Lidz, 2001b; Marelj and Reuland, 2016; Dimitriadis and Everaert, 2014; Reuland and Winter, 2009; Sells et al., 1987; Reuland, 2018; Zec, 1985); other tests are sensitive to syntactic transitivity (here, the behavior of reflexives under causativization). In Turkish, the two classes of tests converge: transitive verbs with pronominal reflexive arguments consistently pattern as dyadic and transitive, and TVRs as monadic and intransitive.²

2.1 Interpretive diagnostics

VP ellipsis treats pronominal and verbal reflexives differently in Turkish: an ellipsis site anteceded by the verb *hazırla-* ‘prepare’ taking a pronominal reflexive object, (6a), licenses both strict and sloppy readings, suggesting that the reflexive pronoun is obligatorily recovered at the ellipsis site. Things work differently in the verbal reflexive counterpart (6b), where only the sloppy reading is available. If the verbal reflexive were secretly a transitive, this contrast should not arise. This differential availability of strict/sloppy ambiguities can also be seen in comparative ellipsis and in the different denials available to assertions modified by *only* (see Dahl 1973; Sauerland 2013; Sportiche 2014).

¹The characteristic syntax of TVRs has allowed us to reliably exclude from this large class verbs that bear non-active morphology but are not syntactically reflexive (e.g. change-of-state unaccusatives and different kinds of motion verbs).

² Turkish judgments originate from the native speaker co-author; we provide examples attested in online sources and published studies where relevant. Regardless of their source, all examples have been confirmed with a approximately 80 Turkish speakers across multiple academic venues. For the data in section 4.4, we initially conducted an informal social media questionnaire with 9 speakers whose judgments have since been confirmed by additional speakers.

- (6) a. Ali kendin-i hazırla-dı, Ayşe de (öyle yap-tı).
 Ali self-ACC prepare-PST Ayşe too so do-PST
 ‘Ali prepared himself, and Ayşe did so too.’ ✓strict ✓sloppy
- b. Ali hazırla-n-dı, Ayşe de (öyle yap-tı).
 Ali prepare-REFL-PST Ayşe too so do-PST
 ‘Ali prepared, and Ayşe did so too.’ ✗strict ✓sloppy

Another diagnostic comes from a type of *deferred reference* (e.g. Fauconnier, 1985; Abusch, 1989; Jackendoff, 1992; Reinhart and Reuland, 1993). Under particular circumstances, instead of picking out the antecedent itself, the reflexive element can instead pick out a contextually salient proxy for the antecedent. Such cases of relaxed identity – termed proxy readings – have been noted to differentiate between different reflexivity strategies cross-linguistically, variously distinguishing between verbal and pronominal reflexives and/or between different kinds of pronominal reflexives (see e.g. Reuland 2018 for a summary, and for analyses see Reuland and Winter 2009; Safir 2004; Raghotham 2024; Paparounas 2023; pronominal reflexives of the so-called SE type seem to not behave uniformly in this test, see Marelj and Reuland 2016 for recent discussion and references). In Turkish, proxy readings clearly distinguish between pronominal and verbal reflexives: the pronominal anaphor *kendi* in (7a) felicitously picks out the proxy in the given context, while the verbal reflexive in (7b) is infelicitous.

- (7) Context: Kivanç sees that his wax statue is getting very dusty, and decides to dust it off.
- a. Kivanç kendin-i silkele-di. b. #Kivanç silkele-n-di.
 Kivanç self-ACC shake-PST Kivanç shake-REFL-PST
 ‘Kivanç shook himself off.’ ‘Kivanç shook (himself) off.’

These facts furnish a further interpretive argument against assimilating TVRs to their pronominal counterparts. Turkish pronominal reflexives clearly involve a transitive syntax: the reflexive DP argument is syntactically distinct from its antecedent. If verbal reflexives similarly involved a transitive syntax with two distinct DPs, we would expect them, too, to give rise to human-acts-on-proxy interpretations, but this is not what we find.

2.2 Syntactic diagnostic: Indirect causatives

Both verbal and pronominal reflexives can be causativized with an *indirect causative* interpretation, which is allowed in Turkish only with predicates that have an external argument (Akkuş 2021, 2023; see also Zimmer 1976). Crucially, in Turkish the case of the causee depends on the transitivity of the causativized base. When a transitive predicate is causativized, the causee bears dative case, whereas in causativization of an unergative intransitive the causee carries accusative case (Lewis 1967; Kornfilt 1997; Göksel and Kerslake 2005; Çetinoğlu et al. 2008; Akkuş 2021, a.m.o).

Predicates with pronominal reflexive arguments pattern as regular transitives: the causee bears dative, (8)-(9). But when a verbal reflexive is causativized, the causee bears accusative case, (10) (cf. Kayne 1975, 203-204, 393 for French *se* reflexives).

- (8) Çocuk kendin-i besle-di. (9) Ebeveynler-i çocuğ- {a / *u} kendin-i besle-t-ti.
 child self-ACC feed-PST parents-3POSS child {DAT / *ACC} self-ACC feed-CAUS-PST
 ‘The child fed himself.’ ‘His parents made the child feed himself.’

- (10) a. *pro* bu ara berbat besle-n-iyor-um.
 this while terrible feed-REFL-PROG-1SG
 ‘I feed (myself) terribly these days.’
 b. [*The speaker complains that his wife doesn’t cook, leading to lots of take-out food.*]
 Merve {ben-i / *ban-a} bu ara berbat besle-n-dir-iyor.
 Merve {I-ACC / *I-DAT} this while terrible feed-REFL-CAUS-PROG
 ‘Merve is making me feed (myself) terribly these days.’ <https://tinyurl.com/mrx2zdes>, 12:40.

That TVRs always appear with an accusative causee is the behavior expected if they are intransitive.

Overall, TVRs behave as monadic predicates interpretively and as intransitives for the syntactic causativization diagnostic; they thus pattern distinctly from pronominal reflexives, which are interpretively dyadic and syntactically transitive. Recall now that the diagnostics we have just deployed are sensitive to valence, but indifferent as to the position of arguments. Turning to this remaining issue for TVRs, we find a striking instance of mixed behavior: the surface subject behaves as a deep object for some syntactic diagnostics, and as a deep subject for others.

3 Internal argumenthood

Various tests show that the surface subject of TVRs originates as an internal argument. These tests distinguish unaccusatives/passives from unergatives/transitives in the language. In this section, TVRs pattern consistently only with unaccusatives and passives, and not with unergatives; whenever TVRs pattern with transitives, their surface subject patterns with transitive objects, not transitive subjects.

3.1 Resultatives

Nominal and adjectival resultatives crosslinguistically serve as a strong indicator of deep objecthood (e.g., Simpson 1983; Levin and Rappaport Hovav 1995). In Turkish, resultatives display the requisite baseline behavior. Firstly, they can only be predicated of direct objects of transitives, as well as surface subjects of passives and unaccusatives (11); crucially, resultatives predicated of subjects of unergatives are systematically impossible, (12).

- (11) a. Leyla masa-yı (ter-temiz) sil-di.
Leyla table-ACC REDUP-clean wipe-PST
‘Leyla wiped the table (completely clean).’ (Turgay 2013:(27a)) *transitive*
- b. Masa (Leyla tarafından) ter-temiz sil-in-di.
table (Leyla by) REDUP-clean wipe-PASS-PST
‘The table was wiped completely clean (by Leyla).’ *passive*
- c. Nehir kas-katı don-du.
river REDUP-solid freeze-PST
‘The river froze solid.’ (Turgay 2013:112,(51a)) *unaccusative*
- (12) *Ceren yop-yorgun koş-tu.
Ceren REDUP-tired run-PST
‘Ceren ran tired.’ (based on Turgay 2013:69, (24b)) *unergative (predicate OK as depictive)*

Secondly, Turkish resultatives are not compatible with unselected objects, (13), or ‘fake reflexives’ of the English type (Turgay, 2013, 69).

- (13) *Leyla demliĝ-i boş iç-ti.
Leyla teapot-ACC empty drink-PST
Intended: ‘Leyla drank the teapot empty.’ (Turgay 2013:89, (27b))

They also cannot be predicated of unprojected elements. In out-of-the-blue contexts, resultatives predicated of missing objects are not licit, (14); they are only licit in *pro*-dropped contexts, (15).

- (14) a. Q: Ne ol-du?
what happen-PST
‘What happened?’
- b. A: Leyla *(masa-yı) ter-temiz sil-di.
Leyla table-ACC REDUP-clean wipe-PST
‘Leyla wiped *(the table) completely clean.’
- (15) a. Q: Leyla masa-ya ne yap-tı?
Leyla table-DAT what do-PST
‘What did Leyla do to the table?’
- b. A: Sadece *pro*_{obj} ter-temiz sil-di.
only REDUP-clean wipe-PST
‘She just wiped it completely clean.’

Overall, the baseline distribution of Turkish resultatives tracks the presence of projected deep objects.

Crucially, resultatives can be freely predicated of the surface subject of verbal reflexives; see (16), with its pronominal reflexive counterpart given in (17). Another example is provided in (18).

- (16) Ter-temiz yıka-n-dı-m.
REDUP-clean wash-REFL-PST-1SG
‘I washed myself clean.’
- (17) Kendi-m-i ter-temiz yıka-dı-m.
self-1SG.POSS-ACC REDUP-clean wash-PST-1SG
‘I washed myself clean.’
- (18) “...Bir sürü Himba insanı görebiliyorsunuz, özellikleri ... (you can see a number of Himba folks, their distinctive feature is ...)”

... kıp-kırmızı-ya boya-n-mış ol-ma-lar-ı.
 ... REDUP-red-DAT paint-REFL-PTCP be-INF-PL-POSS
 ‘them having painted themselves completely red.’ <https://tinyurl.com/5yea9vcn>, 15:45.

TVRs thus have an internal argument position, occupied underlyingly by the surface subject.³

3.2 *Adjectival participles*

A further diagnostic suggesting a low origin for the surface subject of TVRs comes from two types of stative (‘adjectival’) participles in Turkish. Studies of stative participles propose two kinds of analyses of their argument structure. One kind of analysis posits a low origin for the DPs appearing with stative participles, such that a participle is (minimally) a stativized *v*P that includes an internal argument (e.g. [Anagnostopoulou, 2003a](#); [Embick, 2004a](#); [Bešlin, 2022](#)). A different type of analysis posits that stative participles have their arguments introduced externally to the *v*P, on a par with adjectival predication (e.g. [Embick 2023](#), [McIntyre 2013](#), [Paparounas 2023](#), ch. 4). It is crucial in what follows that the participles of Turkish can be independently shown to instantiate the former structure, with a *v*P that includes an internal argument.

One diagnostic showing that Turkish stative passives embed ‘transitive’ (object-ful) *v*Ps comes from passivizable verb-object idioms. Begin with (19), which shows that an idiom like ‘to take the bean out of the mouth’ meaning ‘to disclose a secret’ survives in eventive passives. On standard assumptions about idiom locality, this state of affairs is understandable if the idiomatic reading arises freely with any *v*P that includes the idiom’s fixed parts; as such, the availability of the idiom can be used to diagnose whether the DP *the beans* originated in the same *v*P as the verb *take* in a given case.

- (19) a. Sonunda bakla-yı ağız-dan çıkar-dı.
 finally bean-ACC mouth-ABL take.out-PST
 ‘S/he finally disclosed the secret.’

³Note that the availability of resultatives with TVRs cannot be taken to follow merely from the presence of a theme entailment with TVRs, lest we fail to make the right cut between transitives, passives, unaccusatives and TVRs on the one hand, and examples like (14b) on the other. The resultatives-diagnose-theme-entailments view would also fail to make sense of minimal differences between Turkish and other languages. Consider English and Sason Arabic (data from the native speaker co-author). Reflexive verbs in these languages presumably involve theme entailments just as much as their Turkish counterparts do, yet resultatives are impossible:

- (i) John shaved (*smooth).
(ii) *Nobody else washed Kemal. Instead, on his own...*
#Kemal in-xasal nazzif.
Kemal REFL-wash.PST.3M clean
#‘Kemal washed clean.’ (Sason Arabic)

Whereas the presence/absence of theme entailments does not suffice to explain these differences between Turkish and English/Sason Arabic, the different syntaxes of reflexive verbs in these languages do. Tellingly, independent diagnostics confirm that English *shave*-type verbs have the syntax of unergatives (Reinhart and Siloni, 2004, 2005), and Sason Arabic reflexives do, too. For apparent counterexamples to the direct object restriction, see Biggs 2019a,b on Wechsler 1997.

- b. B20 toplantı-sın-da bakla ağız-dan çıkar-ıl-dı
 B20 meeting-CM-LOC bean mouth-ABL take.out-PASS-PST
 ‘The secret was disclosed at the B20 meeting.’ (Akkuş 2021:253,(533))

Against this background, note that adjectival participles in *-mİş* license the idiomatic reading just like eventive passives do; (20). The participle thus must have its DP argument originating as a deep object.

- (20) Ağız-dan çıkar-ıl-mış bakla-lar huzur-u boz-du.
 mouth-ABL take.out-PASS-PTCP bean-ACC peace-ACC spoil-PST
 ‘The disclosed secrets have upset the peace.’

We accordingly follow much previous work in employing the *-mİş* participle as a low origin diagnostic for Turkish (Nakipoğlu-Demiralp, 1998, 2002; Acartürk, 2005; Gürer, 2014) and languages like it in the relevant respects (Bhatt and Embick, 2004/2017; Alexiadou and Schäfer, 2013).

The pairs in (21) show that the *-mİş* participle is compatible with Roots that typically form unaccusatives (21a), but not unergatives (21b).

- (21) a. {bozul-muş / çürü-müş / eri-miş} yiyecek
 decompose-PTCP / rot-PTCP / melt-PTCP food
 ‘the {decomposed/rotten/melted} food’
 b. {*koş-muş / *yüz-müş / *çalış-mış / *bağır-mış} çocuk
 run-PTCP / swim-PTCP / work-PTCP / shout-PTCP child
 Intended: ‘a {run/swum/worked/shouted} child’ (Nakipoğlu-Demiralp 2002:7a-b)

Moreover, typical transitive-forming Roots allow the formation of participles in *-mİş*:

- (22) a. özensizce kes-il-miş kağıt-lar b. giy-il-miş pantolon
 sloppily cut-PASS-PTCP paper-PL dress-PASS-PTCP pants
 ‘sloppily cut papers’ (Gürer 2014:29a) ‘dressed/put on pants.’

Importantly, TVRs are perfectly compatible with *-mİş*. In (23), we show that it is possible to produce a stative that appears outside the nonactive/reflexive morpheme, and produces a stative-of-a-reflexive-event interpretation. Since we know independently that the *v*Ps that form the input to stativization are those *v*Ps that involve internal arguments, the grammaticality of these examples speaks in favor of a low origin analysis for those *v*Ps that come to be reflexivized before they are stativized.

- (23) a. ört-ün-müş kadın b. eğ-il-miş çocuk
 cover-REFL-PTCP woman bend-REFL-PTCP child
 ‘a self-covered woman’ ‘the self-bowed child’

The same conclusion arises from the language’s other participial formation, formed with the stativizer *-Ik* (Nakipoğlu-Demiralp 1998; Acartürk 2005; Acartürk and Zeyrek 2010; Neu 2026). In

(24a) the *-Ik* participle combines with unaccusative-forming Roots, while it fails to do so in (24b) with unergative-forming ones.

- | | | | | |
|------|----|---|----|---|
| (24) | a. | kır-ık bardak
break-ADJ glass
'the broken glass' | b. | *uyu-k bebek
sleep-ADJ baby
Intended: 'the slept baby' |
|------|----|---|----|---|

The *-Ik* participle is also compatible with anticausative idioms whose sole argument originates in a low position, (25) (the idiom's transitive counterpart can be passivized). Importantly, TVRs can again be the input to the formation of participles (26).

- | | | |
|------|----|---|
| (25) | a. | Adam-ın öd-ü kop-tu.
man-GEN bile-POSS break-PST
'The man got very scared.' (Lit: 'the man's bile broke'; Göksel and Öztürk 2019, 165) |
| | b. | Adam-ın öd-ü kop-uk i-di.
man-GEN bile-POSS break-ADJ COP-PST
'The man was very scared.' (Lit: 'the man's bile was in a broken state.') |
| (26) | a. | Herkes bedence cııl, ruhça ört-ün-ük.
everybody bodily naked soul-wise COVER-REFL-ADJ
'Everybody is bodily naked but soul-wise self-covered.' (Kaynak 2012:75) |
| | b. | Korku-lar-ımız-a sığ-in-ık yaş-ıyor-uz.
fear-PL-1PL.POSS-DAT squeeze-REFL-ADJ live-PROG-1PL
'We live self-refuged in our fears.' (Twitter, @hulyakrt, 9/23/21) |

4 External argumenthood

The diagnostics in the previous section show that the surface subject in TVRs originates low, on a par with the surface subject of passives and unaccusatives. We now demonstrate that the sole argument also passes external argumenthood diagnostics, of the kind that cannot be attributed only to the agentive interpretation of reflexives or to grammatical subjecthood. Putting the diagnostics together reveals that a single argument is associated with two distinct positions inside VoiceP.

Some of the tests examined in this section are traditionally used to contrast transitives and unergatives from unaccusatives in Turkish. Applying these tests to TVRs, we find that their behavior breaks with the patterns established in section 3: here, TVRs pattern with unergatives as opposed to passives/unaccusatives, and their surface subject patterns with the subject, not the object, of transitives.⁴

Crucially, TVRs also pattern distinctly from passives for all relevant tests below. This divergence is important: it clarifies that the fact that reflexives sometimes pattern as external-argument-bearing

⁴In the interest of space, we leave out discussion of a further convergent diagnostic, namely long object movement. Causativization, deployed as an intransitivity diagnostic in section 2.2, in fact constitutes a further external argumenthood test; see discussion near the end of section 4.4.

verbs, and distinctly from unaccusatives, is not attributable solely to the presence of agentive semantics, which reflexives share with passives but not with unaccusatives (cf. [Key 2023](#)). Instead, the initially low surface subject of TVRs must transit through Spec,VoiceP, thereby sharing traits with ‘real’ external arguments for the purposes of the relevant tests. Importantly, we show in section 4.5 that the surface subject of TVRs passes internal and external argumenthood diagnostics in the same example, clarifying that we are indeed dealing with movement, and not structural ambiguity.

4.1 Episodic impersonals

For a majority of Turkish speakers, the addition of the ‘passive’ exponent to any predicate besides transitives with a thematic subject and a structurally case-marked object leads to an *impersonal* construction ([Akkuş 2021](#); [Legate et al. 2020](#)). Unergatives and transitives can form impersonals in both generic and episodic contexts (27)-(28) ([Nakipoğlu-Demiralp 2001](#); [Acartürk 2005](#); [Acartürk and Zeyrek 2010](#); [Akkuş 2021](#); [Legate et al. 2020](#)).

- | | |
|--|---|
| <p>(27) <i>Unergative</i></p> <p>a. Her gece dans ed-il-ir.
every night dance do-IMPERS-AOR
‘People/one dance(s) every night.’</p> <p>b. Konser boyunca uyu-n-du.
concert during sleep-IMPERS-PST
‘People/one slept during the concert.’</p> | <p>(28) <i>Transitive with oblique object</i></p> <p>a. Otobüs-e bin-il-ir.
bus-DAT board-IMPERS-AOR
‘People/one board the bus.’</p> <p>b. Otobüs-e bin-il-di.
bus-DAT board-IMPERS-PST
‘People/one boarded the bus.’</p> |
|--|---|

On the other hand, impersonals of unaccusatives, while felicitous under a habitual reading, (29a), cannot receive an episodic interpretation (29b).

- (29) a. Türkiye-de her gün trafik kaza-lar-ın-da öl-ün-ür.
Turkey-LOC every day traffic accident-PL-CM-LOC die-IMPERS-AOR
‘In Turkey people die in traffic accidents every day.’ ([Nakipoğlu-Demiralp 2001:140,\(22b\)](#))
- b. *Dün burada öl-ün-dü.
yesterday here die-IMPERS-PST
Intended: ‘People/one died here yesterday.’ (Unaccusative)

As noted in [Akkuş 2021](#) and [Legate et al. 2020](#), impersonals involve an unpronounced impersonal pronoun (IMP) filling an argument position – they thus lack argument ‘demotion’. The position occupied by the impersonal pronoun can be that of the thematic subject, as in (27)-(28), or the thematic object (29). Crucially, the availability of an episodic interpretation is modulated by the syntactic position of IMP: episodic interpretations are allowed with IMPs in Spec,VoiceP, but disallowed with IMPs merged as internal arguments.⁵

⁵Romance languages show similar effects, inviting identical analyses ([Ordóñez, 2021, 155](#)).

TVRs pattern with unergatives and transitives in being compatible with both habitual and episodic aspects. Consider the examples in (30) with episodic aspect:

- (30) a. Bu çim-ler-e uza-n-ıl-mış.
this grass-PL-DAT lie.down-REFL-IMPERS-PST
‘Apparently people/one laid (themselves) down on the grass.’
b. Büyük-ler-in ön-ün-de saygı-yla eğ-il-in-di.
elder-PL-GEN front-POSS-LOC respect-with bend-REFL-IMPERS-PST
‘People/one bowed respectfully in front of the elder people.’

To repeat, verbal reflexives pattern with ‘active’ structures for the purposes of impersonals, in which a syntactically projected argument is found in Spec,VoiceP.

4.2 Agent nominalization

The availability of agent/external argument nominalization has sometimes been used as a diagnostic of the argument structure of verbal reflexives; see e.g. Reinhart and Siloni (2005); Alexiadou and Schäfer (2013); Amato (2024); Paparounas (2023). Turkish provides the appropriate baseline to apply the test: agent nominalization is sensitive to the unergative/unaccusative distinction, independently of verbal reflexives. Thus, Roots that typically form unergatives/transitives freely form agent nominals in *-IcI*, (31a). Unaccusative-forming Roots, however, resist agent nominalization, (31b) (see Nakipoğlu-Demiralp 1998; Acartürk 2005). (31c) is a particularly probative example: it shows that a typical unaccusative-forming Root, in this case $\sqrt{\text{FINISH}}$, can only yield an agent nominal if it is first transitivized.

- | | | | |
|------|------------------------------------|---|--|
| (31) | a. koş-ucu
run-NMLZ
‘runner’ | b. *düş-ücü
fall-NMLZ
Int: ‘faller’ | c. bit-*(ir)-ici
finish-CAUS-NMLZ
‘finisher’ |
|------|------------------------------------|---|--|

Consider now the attested examples in (32) showing that TVRs can be the input to agent nominalization; crucially, these are *not* Root nominals, witness the surfacing of Voice morphology between the Root and the nominalizer:

- (32) a. “... eğer padişahlığ-a er-ecek ol-ur-sa-n haram-dan sak-in-ıcı ol.”
if sultanate-DAT reach-FUT be-AOR-COND-2SG sin-ABL save-REFL-NMLZ be
‘If you reach the sultanate, be an avoider of sin [lit. a self-protector-from sinning].’ (Şahin 2019:79)
b. Putin Hindistan-da. Seri sar-ıl-ıcı Modi tabii on-u da kucakla-dı.
Putin India-LOC serial wrap-REFL-NMLZ Modi of.course he-ACC also embrace-PST
‘Putin is in India. The serial hugger (lit: self-wrapper) Modi of course gave him a hug too.’ (Twitter, @OmerFarukGorcin, 10/5/18)

Previous literature has sometimes claimed that agent nominalizations of TVRs are not well-formed (e.g., [Gündoğdu 2017](#)). As far as we can tell, such claims result from individual examples having been judged in isolation and without context. We have provided attested examples here, additionally judged as grammatical by our consultants, to help control for the kinds of pragmatic factors that are known to affect the acceptability of derived nouns more generally. Firstly, the agent nominal must be afforded a specialized usage in lexicosemantic space; this is often achieved via modification (e.g. *serial hugger* in (32b); cf. e.g. [Embick and Marantz 2008](#) on English *#stealer* versus *base-stealer*). Secondly, agent nominals must denote entities that are *name-worthy* in a sense familiar from the literature on noun incorporation (see e.g. [Carlson, 2006](#), 10ff) and implicit indefinites (see e.g. [Martí, 2015](#)), describing activities institutionalized enough to be worthy of a label (e.g. *dile-n-[i]ci* ‘beggar (lit: self-wisher)’ is a well-established agent nominal constructed from a verbal reflexive). TVRs then can, in fact, undergo agent nominalization in the general case, though judgments on individual formations are often modulated by grammar-external conditions of appropriateness of use.

The capacity to form agent nominals is thus another trait that TVRs share with unergatives, deviating from passives and their ilk. Various examples with agent nominals further highlight the dissociation between passives and reflexives. For example, the verb *sar-ıl* ‘wrap-NACT’ in (32b) is ambiguous between passive (‘to be wrapped around’) and reflexive (‘to wrap oneself around’) interpretations in finite contexts. But the addition of the agent nominalizer *-Ici* leaves behind only the reflexive interpretation: the form *sar-ıl-ıci* cannot mean ‘one who gets hugged/wrapped around’, but only ‘one who hugs/wraps oneself around (someone)’.

4.3 Adverbial gerundive -ArAk

In this section we consider adverbial gerundives with the verbal suffix *-ArAk* ([Özkaragöz 1980](#), [Knecht 1985](#), [Biktimir 1986](#), [Kornfilt 1997](#), [Legate et al. 2020](#), [Akkuş 2021](#), [Akkuş and Paparounas 2024](#)). *-ArAk* clauses display intricate matching restrictions that place verbal reflexives in the same natural class as active transitives/unergatives, distinct from passives or unaccusatives. Briefly, *-ArAk* gerunds necessitate that the matrix and gerundive clause match in voice, in the status of the subject as underlying or derived, or a combination of both. Table 1 summarizes the relevant baseline patterns.⁶

Verbal reflexives behave like unergatives/transitives, (33a-33b), in being able to combine with each of these classes but not with unaccusatives, (33c).

⁶Most of the baseline generalizations on *-ArAk* are distributed across various studies; the behavior of TVRs in these gerunds was first noted in [Akkuş and Paparounas 2024](#), replicated here. In this paper we only provide examples with TVRs in the interest of space.

Transitive/unergative + transitive/unergative	✓
Unaccusative + unaccusative	✓
Unergative + unaccusative	✗
Passive + passive	✓
Passive + transitive/unergative	✗
Passive + unaccusative	✗
Verbal reflexive + transitive/unergative	✓
Verbal reflexive + unaccusative	✗
Verbal reflexive + passive	✗

Table 1: *Patterns of combinations with -ArAk*

- (33) a. Kız [söyle-n-erek] yürü-dü.
 girl say-REFL-ARAK walk-PST
 ‘The girl walked (while) complaining.’ (reflexive + unergative; Nakipoğlu-Demiralp 2002, 13c)
- b. Adam [söyle-n-erek] bulaşık-lar-ı yıka-dı.
 man say-REFL-ARAK dish-PL-ACC wash-PST
 ‘The man did the dishes (while) complaining.’ (reflexive + transitive)
- c. *Adam [buna-yarak] yıka-n-dı.
 man go.senile-ARAK wash-REFL-PST
 ‘The man washed (while) going senile.’
 The man was washed (while) going senile. (unaccusative + reflexive)

Moreover, reflexives are not compatible with passives, clarifying that the mere presence of an agentive entailment in reflexives does not suffice to explain why reflexives do not combine with unaccusatives.

- (34) a. *Manken [giy-in-erek] öp-ül-dü.
 model dress.up-REFL-ARAK kiss-PASS-PST
 ‘The model was kissed (while s/he was) getting dressed up.’
- b. *Çocuk [okşa-n-arak] söyle-n-di.
 child caress-PASS-ARAK say-REFL-PST
 ‘The child complained (while s/he was) being caressed.’

In terms of *-ArAk* clauses then, reflexives pattern with unergatives and transitives (which have a filled Spec,VoiceP), and distinctly from both unaccusatives (which lack a thematic Voice layer altogether), and from passives (which have a thematic Voice, albeit one lacking a specifier).

4.4 Non-passivization

Verbal reflexives may not be passivized in Turkish, and instead result in impersonals (Legate et al. 2020; Akkuş 2021) - a fact again expected if they behave, for the purposes of this diagnostic, like active Voice with an argument occupying Spec,VoiceP, and not passive Voice (cp. e.g. Schäfer, 2012).

Sequences of Refl+‘Pass’ (observed in the prior literature, e.g., Göksel 1993) are ungrammatical when a *by*-phrase is added, (35). The addition of the ‘passive’ morpheme leads to an *impersonal* interpretation, and we accordingly translate the relevant examples as ‘People/one self-Verbed’.

- (35) a. Dün dere-de (*adamlar tarafından) yıka-n-ıl-dı.
yesterday river-LOC men by wash-REFL-IMPERS-PST
‘People/one self-washed in the river yesterday (*by some men).’
b. Sıcak ol-duğ-u için (*tatilciler tarafından) soy-un-ul-du.
hot be-NMLZ-POSS for vacationers by undress-REFL-IMPERS-PST
‘People/one self-undressed (*by the vacationers) since it was hot.’

Further confirmation of the impersonal reading comes from the fact that TVRs with the ‘passive’ morpheme require a human interpretation, (36), and that only 3SG agreement is allowed, (37).

- (36) a. Bu orman-da {çocuk-lar / fil-ler } dere-de yıka-n-ır-lar.
this forest-LOC {child-PL / elephant-PL } river-LOC wash-REFL-AOR-PL
‘In this forest, {children / elephants} wash (themselves) in the river.’
b. Bu orman-da dere-de yıka-n-ıl-ır.
this forest-LOC river-LOC wash-REFL-IMPERS-AOR
‘In this forest, {people / #elephants} wash (themselves) in the river.’
(37) *Hamam-da yıka-n-ıl-dı-m.
Turkish bath-LOC wash-REFL-PASS-PST-1SG
Intended: ‘Self-washing was done by me in the Turkish bath.’

The non-passivizability of verbal reflexives is instructive in indicating the presence of an argument in Spec,VoiceP. The crucial observation is that verbal reflexives behave in the same way as impersonals. Following the analysis of impersonals in Akkuş 2021 and Legate et al. 2020 (cf. section 4.1 for a summary), the absence of passivization straightforwardly follows if the IMP occupies the specifier of VoiceP, such that TVRs pattern like active transitives or unergatives for this test.

The patterns in this section also highlight that the behavior of TVRs in causatives, noted in section 2.2, provides an additional external argumenthood test. Since indirect causatives embedding verbal reflexives are only possible with structures that have a filled Spec,VoiceP in Turkish (see Akkuş and Paparounas 2024), the fact that verbal reflexives form indirect causatives yields another argument that their initially low surface subject transits through Spec,VoiceP.

4.5 External-Internal properties combined

The surface subject of TVRs can simultaneously pass internal and external argument diagnostics in the same example. In (38a), the reading of an arbitrary human impersonal in the episodic context is reserved for external arguments; but the resultative points to the presence of an internal argument.

Likewise, in (38b), causativization signals the presence of an argument in embedded Spec,VoiceP (see sections 2.2 and 4.4), with the resultative again suggesting the presence of a deep object.

- (38) a. Dün bu dere-de ter-temiz yıka-n-ıl-dı.
 yesterday this river-LOC REDUP-clean wash-REFL-IMPERS-PST
 ‘People/one washed clean yesterday.’
 b. *pro* [çocuğ-u ter-temiz yıka-n]-dır-dı-m.
 [child-ACC REDUP-clean wash-REFL]-CAUS-PST-1SG
 ‘I made [the child wash clean].’

Such combinations call for a structure that can accommodate the internal and external argument properties of the same argument in the same example; they thus rule out the possibility that TVRs are ambiguous between unergative and unaccusative structures. Labiality of this kind of course arises quite generally elsewhere (for unergative/unaccusative alternations, see e.g. Perlmutter 1978; Burzio 1981, 1986; Sorace 2000; Krejci 2020; Holisky 1987; Neu 2026) and arguably with reflexives in other languages (see esp. McGinnis 2022 for *se* reflexives); but it does not suffice to explain our data.

Besides suggesting the necessity of a movement analysis as opposed to mere structural ambiguity, examples like (38) reinforce a point already evident in our discussion up to this point: different phenomena can be sensitive to distinct positions forming part of the same movement chain. In (38), it suffices for the licensing of the resultative that the surface subject originates as a deep object. For the purposes of impersonals and causatives, by contrast, it suffices that the same argument ended up occupying Spec,VoiceP on its way to becoming the grammatical subject. This type of differential sensitivity of particular phenomena to particular positions along a movement chain is familiar from other domains (see e.g. McCloskey 1997 on the base position of ‘subjects’; Levin and Rappaport Hovav 1995, 17ff for the surface subjects of unaccusatives; and Kastner 2017, 17ff for relevant discussion with specific reference to reflexives).⁷

5 Analysis: Reflexive Voice and Movement

5.1 Syntax

The Turkish facts we have surveyed up to this point call for a mechanism of reflexivization capable of deriving a reflexive interpretation from a single-argument structure, one where the single argument in

⁷An anonymous reviewer suggests examining *theme unergatives* (in the sense of Reinhart 2002; Levin and Rappaport Hovav 1995 and related work) as a way of dissociating thematic role from syntactic position, and thereby confirming that our positional diagnostics are sensitive to the former and not simply the latter (cf. footnote 3). In Turkish, such predicates (e.g. *kokmak* ‘stink’, *parıldamak* ‘sparkle’, *ciyaklamak* ‘squeal’) pattern with regular unergatives for every conceivable diagnostic (e.g. they freely form episodic impersonals; they never form adjectival participles; they pattern like other unergatives in *-ArAk* gerundives). If the sole argument of these verbs is a theme, then this data confirms that our diagnostics track syntactic position rather than thematic role. We thank the reviewer for raising this point.

question has a basic position VP-internally and comes to occupy a derived position VoiceP-internally before attaining its ultimate position as grammatical subject.

Section 2 showed that TVRs are not transitive verbs. We must thus derive a reflexive interpretation from an intransitive structure, without resorting to a (null) reflexive pronoun in the syntactic structure of these verbs. In unergative analyses of reflexives – where the sole argument is base-generated externally to the *v*P (see e.g., Chierchia 2004; Bruening 2006; Reinhart and Siloni 2004, 2005; Labelle 2008; Jo 2019) – this is an easy task: with an appropriately typed *v*P, a Voice head could effectively ‘pull out’ the unsaturated theme role and identify it with a distinct element. But deriving a reflexive denotation from a basic unaccusative structure is an underappreciated challenge, since identifying the entity already mapped to theme with a distinct element in a non-stipulative way is non-trivial; we refer the reader to Paparounas (2023, 115ff) for a more detailed exposition of the issue. Existing unaccusative analyses, when explicit on the semantics, resort to exceptional placements of lambda abstracts (see e.g. Spathas et al. 2015, cf. Barker 2007), departures from standard assumptions on the interpretation of movement (e.g. Kastner 2017), or direct identification of thematic roles (Paparounas, 2023; Oikonomou and Alexiadou, 2022), a move unclear in its formal details.

We note also that in Turkish, as in other languages with similar voice systems, verbal reflexivization seems strongly tied to grammatical voice: TVRs surface with a particular voice morphology, and that morphologically (and syntactically) active verbs never yield verbal reflexives, as noted in §1.

We accordingly propose to link reflexivity to the encoding of grammatical voice, which we take to be carried out in large part by the head Voice, which determines the properties of the external argument. We posit a reflexive variant of Voice (following Labelle 2008; McGinnis 2022; Raghotham 2022; Oikonomou and Alexiadou 2022; Paparounas 2023; Kallulli and Roberts 2025; cf. Spathas et al. 2015; Alexiadou 2014; and for partially related ideas in anaphor binding, see e.g. Ahn 2015; Kratzer 2009; Paparounas and Akkuş 2024). The move in these works is to treat ‘verbal’ reflexivity as a kind of external argument introduction; it is this type of analysis we propose for Turkish. Effectively, if active Voice introduces an agent function as well as a DP that saturates it, and passive Voice an existentially closed agent function, reflexivity can also be understood as a kind of agent introduction, one where the agent role ultimately comes to be identified with a distinct role.

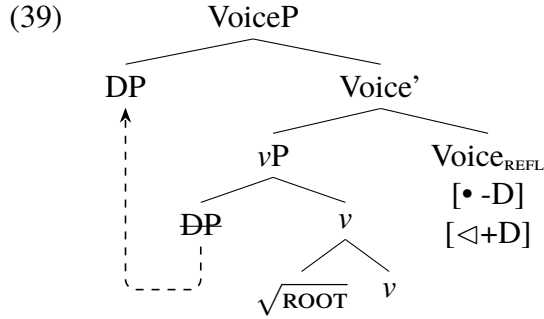
Recall from section 1 that the set of Roots capable of forming TVRs is neither small nor lexicosemantically circumscribed. In our analysis, selectional restrictions must guarantee that the reflexivizing Voice head will appear only with those Roots, or Root+*v* combinations, capable of forming verbal reflexives in Turkish; in this it resembles other Voice heads (see section 1 as well as Alexiadou et al.

2006; Schäfer 2008a; Alexiadou and Schäfer 2013). There are different ways of implementing this selectional relationship: for instance, it could be that the identity of the Root is visible for selection at the *v*P level (see Merchant 2019 for categorizers and Roots operating jointly in selection), or the relevant Roots could bear a diacritic feature visible at the *v*P level. Since the choice between such alternatives would leave our analysis otherwise unaffected, we note only that the existence of such restrictions (however implemented) is unremarkable in light of other domains of argument structure. It is not uncommon cross-linguistically for there to be lexical restrictions on the formation of causatives (e.g. Pylkkänen, 2008, 107ff); anticausatives (e.g. Alexiadou et al., 2006; Schäfer, 2008a); or passives (e.g. Doron, 2003; Alexiadou and Doron, 2012), with the set of Roots that do or do not appear in these structures typically not characterized solely on a lexicosemantic basis. Our contention here is simply that TVRs are no different from these other voice alternations.

The basic syntax of Voice_{REFL} is characterized by its inability to introduce a DP. Following the notation in Müller (2010) and subsequent work, we notate this restriction as a feature [$\bullet$ -D] which prohibits external Merge of Voice_{REFL} with elements of category D. Voice_{REFL} shares this feature with other nonactive Voice heads in the sense of Alexiadou et al. 2015 (see also Schäfer 2008a; Legate 2014; Legate et al. 2020): the Voice heads building passives and (marked) unaccusatives also bear [$\bullet$ -D], and it is this natural class that leads to the emergence of non-active morphology across these categories in Turkish (see Key 2024 for Turkish, and for broader discussion see Alexiadou et al. 2015; Paparounas 2023; Embick 1997, 2004b; Kastner 2016; Oikonomou and Alexiadou 2022).

Though Voice_{REFL} resembles other non-active Voice heads in its inability to be Externally Merged with a DP, it differs in demanding to be combined with a DP via Internal Merge. Voice_{REFL} bears the trigger of the movement step diagnosed in sections 3-4: we take this to be an Internal-Merge-triggering feature [$\triangleleft$ D] (following Hewett 2023), as in (39) below. This is an EPP feature posited independently in work on different phenomena (see e.g. Hopperdietzel 2024 for Mandarin, Deal 2019 for Nez Perce, Lee 2024 for Korean, Sundaresan 2012 for Tamil; cf. Ahn 2015 for English anaphors; and see discussion of edge features elsewhere including Chomsky 2008; Abels 2012; Georgi 2017).

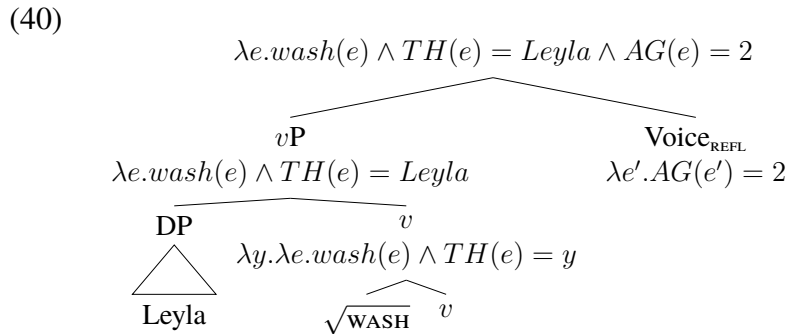
⁷A selection-free alternative to the issue would posit that the grammar in fact generates all possible combinations, with aberrant ones deemed impossible only at the level of world knowledge (e.g. Marantz, 1997; Borer, 2005). Since we have argued that encyclopedic content alone (either at the level of lexical semantics or at that of world knowledge) does not always predict which Root will and will not form a TVR, this route is not available to us.



5.2 Interpretation

With this basic syntax in place, we propose an account of how reflexivization comes about on Voice; in doing so, we develop ideas on the precise interpretation of Reflexive Voice originally suggested to us by Kyle Johnson (p.c.), to whom we are indebted. We take thematic roles to be functions from events to individuals in a neo-Davidsonian event semantics (see e.g. [Parsons, 1990](#)), introduced by particular functional heads at LF, as is common in recent argument structure literature (e.g. [Myler 2016](#); [Wood 2015](#); [Bruening 2013](#); [Legate 2014](#); [Alexiadou et al. 2015](#); [Schäfer 2008b](#); [Borer 2005](#); for antecedents see [Chomsky 1993](#), [Baker 1997](#), 121-122, [Reinhart 2016](#), 3). The theory we assume thus has no thematic positions in the strict sense, contrasting with a more traditional conception of θ -roles as entities in the syntax (or the lexicon-syntax mapping, [Chomsky 1981](#)).

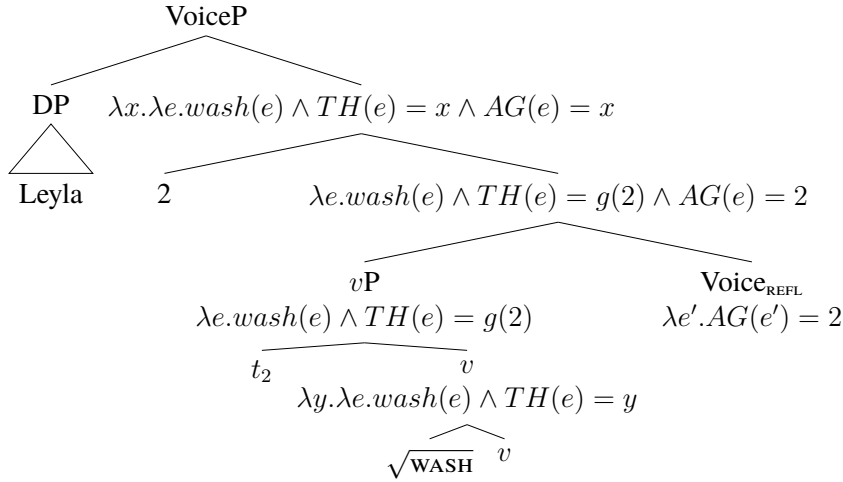
In (40), the denotation of $\text{Voice}_{\text{REFL}}$ introduces an Agent function, like other (thematic) Voice heads; but it also introduces a free variable, represented as a numerical index feature 2 ([Heim and Kratzer 1998](#), [Kratzer 2009](#)). This free variable saturates the Agent function, providing an $\langle s, t \rangle$ denotation for the Voice head that can combine with the vP meaning via Predicate Modification.



For the denotation in (40) to ultimately be well-formed, the index 2 will have to be assigned an interpretation. We propose that this index can only be interpreted under semantic binding, made possible by movement of the internal argument. Our analysis is inspired by the Binder Index Evaluation Rule of [Büring \(2005\)](#), where quantified NPs come with an index demanding that pronouns c-commanded by such NPs and bearing the same index are necessarily bound pronouns. Our approach similarly suggests that the index 2 on $\text{Voice}_{\text{REFL}}$ is to be interpreted as a bound variable (see

Altshuler et al. 2019 for a concrete separation of indices into discourse-related and trace-related; also see Kratzer 2009). Once *Leyla* moves, it binds both its trace and the free variable introduced on Voice.

(41)



In (41), we follow the convention in Heim and Kratzer (1998), where the landing site of movement involves an index that triggers lambda abstraction. The lambda abstract triggered by the index 2 comes to bind both the trace bearing that index and the free variable on Voice.⁸

Note that representations like (41) are LF trees: we are assuming that they result from a set of principles mapping syntactic movement chains to configurations of semantic binding (see e.g. Buring 2005, and for brief discussion linked directly to verbal reflexives, see Reuland 2018, 90-91). In particular, we are assuming that LF translates lower copies of movement into variables (or larger structures embedding variables; see e.g. Fox 1999; Sauerland 2004; Poole 2024), and that lambda-abstraction-triggering indices are effectively sprouted at the landing site of movement as just mentioned, yielding configurations like (41). In other words, though elsewhere adhering to the program in Heim and Kratzer (1998), we treat a substantial part of the machinery developed therein as obtaining at LF; in particular, we do not take indices to be present in the syntax proper (Chomsky, 1995; Reuland, 2011).⁹

5.3 Consequences and comparisons

Our approach successfully captures a range of important properties of TVRs.

⁸Situations like this are by no means uncommon. In examples like *Every student/Mary seems to herself to have passed the exam*, the embedded subject raises to matrix grammatical subject in a way that allows it to bind both its trace and the index on the reflexive pronoun located in the matrix clause (see e.g. Reinhart 2006, 165ff, Reuland 2011, 348; this is usually discussed with reference to quantified expressions, but naturally all A-movement should have this property).

⁹*Modulo* broader architectural consequences not specific to our analysis, our account could in principle be translated into a framework where indices and variables are assigned a narrow-syntactic life (see e.g. Bruening 2021 for one such approach to certain implicit complements).

Firstly, it successfully derives a reflexive interpretation from an unaccusative/passive-like structure; a non-trivial task, as outlined above. Secondly, the approach successfully captures the interpretive properties of TVRs, diagnosed in §2.1. The structure in (41) includes only a single syntactic argument, the DP *Leyla*; the index 2, like the trace of *Leyla*, is non-distinct from *Leyla* for the purposes of the interpretive phenomena. In particular, (41) does not employ an anaphoric argument to saturate a thematic role, as the derivation of a transitive verb with a pronominal reflexive would; rather, the inclusion of the free variable derives the effect of reflexivizing the agent role, once semantic binding of this variable takes place. The assumption that the free variable can be interpreted only under binding, and not coreference, derives the interpretive asymmetries between pronominal and verbal reflexives. Whereas Turkish pronominal reflexives can be interpreted either via coreference or via binding under ellipsis, the free variable on Voice is only interpretable under binding, and thus fails to allow both strict and sloppy readings; pronominal reflexive arguments are sufficiently distinct from their antecedents to license proxy readings, whereas the free variable is not (see Paparounas 2023; Raghotham 2022; Reuland and Winter 2009, ch. 3 for relevant discussion); and other interpretive limitations characteristic of TVRs but not shown here for reasons of space also follow (e.g. obligatoriness of *de dicto* readings with attitude verbs). Overall, the configuration we postulate builds a unary predicate out of the verb phrase by means of how the index on reflexive Voice is interpreted (cp. the notion of syntactic bundling in Reinhart and Siloni 2005); in all relevant respects, the free variable resembles the interpretation of lower copies of movement chains, which themselves never license strict/sloppy ambiguities or partial identity readings.

Thirdly, the association of reflexivity with Voice straightforwardly captures the fact that reflexivization in Turkish goes hand in hand with a particular kind of voice morphology. The fact that reflexives are voice-selective, appearing only with nonactive morphology, would be mysterious if reflexivization were orthogonal to Voice: concretely, we would have no explanation for why a morphologically active verbal form cannot be interpreted reflexively.¹⁰

Further contrasts between pronominal reflexives and verbal reflexives follow. For instance, the Turkish pronominal reflexive can be bound long-distance, as in (42) (Kornfilt, 1997, 2001), but TVRs disallow this possibility, (43). Since theta-role identification takes place locally on the reflexive Voice head, the impossibility of a long-distance interpretation of TVRs naturally follows on our analysis.

¹⁰Not all verbal reflexives syncretize with passives in Turkish (Kornfilt, 1997; Göksel, 1993; Key, 2023, 2024). As Key notes, it is *ground* reflexives that are realized with a dedicated exponent not shared with the passive, see section 6.

- (42) Çocuk [anne-si kendi-ni yıka-yacak diye] duy-du.
 boy mother-his self-ACC wash-FUT that hear-PST
 ‘The boy_i heard that his mother_k will wash him_i/herself_k.’
- (43) Çocuk [anne-si yıka-n-acak diye] duy-du.
 boy mother-his wash-REFL-FUT that hear-PST
 ‘The boy_i heard that his mother_k will wash *him_i/herself_k.’

We conclude this section by comparing our analysis of TVRs and existing analyses of different kinds of reflexive verbs that capitalize on movement in different ways. The shape of the semantic derivation we propose is clearly reminiscent of the standard analysis of Romance clitic reflexives developed in e.g., [Marantz 1984](#); [Kayne 1988](#); [Pesetsky 1995](#); [McGinnis 2004](#): in this type of analysis, movement of a deep object DP to the grammatical subject position across a pronominal clitic argument introduced in the specifier of Voice ensures Condition A-compliant syntactic binding of the clitic anaphor and produces thematic role identification. Ultimately, the analyses differ. In Turkish, the crucial movement can be shown in Turkish to be VoiceP-internal, distinct from the ordinary promotion to grammatical subject undertaken by the deep object of passives and unaccusatives without the intermediate movement step that characterizes TVRs. Moreover, since there is no pronominal reflexive argument in Turkish (see section 2.1, our analysis posits no syntactic binding, but only semantic binding of the free variable contributed by Voice.

Prima facie, our analysis is closer in spirit to the proposals for movement-driven reflexivization in [Hornstein \(2001\)](#); [Boeckx et al. \(2008\)](#). The analysis of English *shave*-type verbs in [Hornstein \(2001, 41\)](#), schematized in (44), bears an obvious initial resemblance to our analysis of Turkish.

- (44) [_{TP} John [_{VOICEP} John [_{VP} shaved John]]].
 ^-----^-----
 (Diagram showing movement of John from the object position to the subject position of the VoiceP)

In substance, the specifics of our analysis ultimately diverge from these existing proposals in important ways. The theory of construal in [Hornstein \(2001\)](#), further specified for different kinds of reflexive dependencies in [Boeckx et al. \(2008\)](#), involves saturation of two distinct roles by one and the same argument (cf. [Grohmann 2003](#), ch. 3). Under the traditional view of thematic roles presupposed in these works, this necessitates abandoning (or relaxing) the *Theta Criterion* ([Chomsky 1981](#); see [Collins 2024](#) for a contemporary view). In our analysis, there is no technical sense in which the same entity saturates two separate thematic functions. The displaced internal argument ends up semantically binding the free variable on Voice, and this step leads to identification; while the final denotation ends up having the same definite description appearing as the argument of both Theme and Agent, this result has been arrived at through semantic binding, not by the same element saturating

one function in its base position and one in its landing site. More broadly, since we do not begin from a conception of thematic roles as syntactic entities, compliance with the Theta Criterion is simply not at stake in the first place (Heim and Kratzer, 1998, 51). The view of thematic interpretation we assume – whereby thematic roles are simply neo-Davidsonian functions introduced at LF – in fact predicts different kinds of misalignments between syntactic structure and thematic interpretation that are not compatible with strict biuniqueness between syntactic arguments and thematic roles (see e.g. Legate 2014; Bruening 2013 on implicit agents in passives, and Wood 2015; Myler 2016 on so-called *delayed saturation*).

Our analysis of Turkish also differs in scope from that in Hornstein 2001. The theory Hornstein 2001 is intended as a general theory of referential dependency, covering not just cases of reflexivization without an obvious pronominal source, but also cases of antecedent-anaphor binding (where the anaphor is analyzed as the realization of a lower copy of the mobile antecedent) and of control. This theory further has cross-linguistic ambitions, intended as an account of construal across languages.

Our claims are more modest. We do not wish to claim that movement-based identification is the only strategy to build verbal reflexives cross-linguistically; put simply, we consider the Turkish case noteworthy precisely because it seems to clearly evidence a type of derivation at best underevidenced, and at worst counterevidenced, elsewhere. The derivation of English *shave* in (44) is a case in point. If thematic role identification in English proceeded as it does in Turkish, namely, by movement within the VoiceP, then we would expect the surface subject of such verbs in English to pass the same kinds of low origin diagnostics that its Turkish counterpart does. But this is not the case; for instance, English reflexive verbs do not license resultatives (see fn. 3, and cf. Reinhart and Siloni 2005 for more evidence in favor of a basic unergative syntax for English reflexive verbs).

This is not to say that the individual pieces of our analysis cannot be expected to be found in other languages; in the next section we offer some first directions for predictive avenues that our analysis opens up. But we do not claim that our analysis should be ported over wholesale to other structures that have been discussed under the label ‘verbal reflexive’. For different views on the extent to which the syntax of categories described under this label should be expected to be uniform crosslinguistically, see Kallulli and Roberts (2025); Reinhart and Siloni (2005); Marelj and Reuland (2016); Embick (2004b); Alexiadou and Schäfer (2013); cf. Legate (2021) on the non-uniformity of ‘passives’.

Our discussion also explicitly prohibits us from extending the movement-based analysis to pronominal reflexives, again *pace* Hornstein (2001). If we were to assign the same derivation to verbal and pronominal reflexives in Turkish, the many differences between the two that we have highlighted

would go unexplained; this specific concern is superimposed on the many independent concerns with positing movement as *the* way to relate reflexive pronouns and their antecedents, some familiar from parallel critiques of the movement theory of control (e.g., Landau 2003; Bobaljik and Landau 2009; Sigurðsson 2008; Wood 2012).

A final movement-based view of reflexivization worth comparing with our own is the analysis of Hebrew reflexive verbs in Kastner 2017. This analysis adjoins to nonactive Voice an agentive modifier $\sqrt{\text{ACTION}}$; the agent role contributed by this modifier is saturated by movement of the internal argument to a higher position, taken to be Spec,TP. The way the analysis is implemented requires a non-standard view of the interpretation of movement, one where the moving element is interpreted as a definite description in both its base position and landing site (see Kastner 2017, 14-15; Kastner does not directly diagnose the placement of the surface subject, but nonactive morphology is indeed suggestive of its low origin). We consider this view to be in need of further elaboration, since it does not capture the interpretation of movement chains elsewhere. Our analysis does not require a modifier of the $\sqrt{\text{ACTION}}$ type.

6 Conclusion

We have shown that the surface subject (and sole argument) of Turkish verbal reflexives occupies two positions within the VoiceP, patterning as an internal argument for some diagnostics and as an external argument for others, suggesting movement from the former to the latter position. We have taken it not to be an accident that we find movement between the position canonically associated with the theme role and that associated with the agent role; and that this type of movement is attested only in verbal reflexives, which involve the identification of these two roles, but not in other structures such as unaccusatives or passives. We have proposed a movement-triggering variant of the Reflexive Voice head, coupled with a view of its semantics that succeeds in deriving a reflexive interpretation from a single-argument syntax.

Throughout, we have argued for the correctness of this approach for the case of Turkish. Though we cannot exhaustively examine its cross-linguistic applicability, we conclude by outlining certain new avenues it opens up for cross-linguistic inquiry.

These predictions involve the thematic restrictions that characterize verbal reflexives. Cross-linguistically, structures described as verbal reflexives tend to be agent-oriented (see Reuland 2018 for summary). An analysis that ties verbal reflexivization to the introduction of agents in syntax by means of Reflexive Voice derives such restrictions straightforwardly (Alexiadou, 2014; Spathas et al., 2015; McGinnis, 2022; Paparounas, 2023): as long as Voice heads more generally are taken to introduce

Agents, the reflexive flavor will be capable of only deriving agent-oriented reflexives. We believe the more specific view offered here, where Reflexive Voice can be combined with movement, raises the possibility of explaining more finer-grained restrictions concerning the range of roles identified with the agent in Voice-based reflexivization.

On our analysis, Voice-based reflexivization is effected jointly by the free variable contributed by the Voice_{REFL} head, and by syntactic movement of the internal argument that yields binding of that free variable at LF. This view makes an interesting prediction: if to be reflexivized via Voice is to A-move in order to bind the free variable on Voice_{REFL}, then independent restrictions on A-movement will constrain the range of reflexivization possibilities. In particular, only elements independently capable of undergoing A-movement will be able to participate in Voice-based reflexivization. In particular, the account developed thus far raises two expectations:

- (45) a. In a given language $\mathcal{L}$ allowing Voice-reflexivization, if a DP α originating in a position canonically associated with a given thematic role θ can undergo A-movement to Spec,VoiceP, then θ can be identified with the role Agent, all things being equal.
- b. If A-movement from a position canonically associated with a given thematic role θ is independently impossible in $\mathcal{L}$, then θ cannot be Voice-reflexivized with Agents in $\mathcal{L}$.

Here, we limit ourselves to two indicative initial looks into these expectations. Examining whether they are borne out more widely is one way to test the cross-linguistic applicability of our analysis.

A first data point is Turkish-internal. Key (2023, 2024) discusses two types of Turkish verbal reflexives differing as to which role is identified with the agent: figure (i.e. *theme*) or ground (an umbrella term for the roles *location*, *beneficiary* and *experiencer*). The sole argument in these reflexives is introduced in distinct projections: *vP* for figure reflexives and *ApplP* for ground reflexives (see also fn. 10). This bifurcation has the benefit of capturing the existence of transitive ground reflexives in Turkish, which involve a separate theme/figure argument separate from the ground, as in (46).

- (46) Davetsiz misafir-ler sandviç-ler-i tı-ın-di-lar.
uninvited guest-PL sandwich-PL-ACC stuff-REFL-PST-PL
'The uninvited guests stuffed themselves with sandwiches.'

Transitive ground reflexives can be incorporated into our analysis as in (47). This structure contains a separate figure DP, making for a transitive structure; the surface subject originates in the specifier of Appl, and movement to spec, Voice_{REFL}P derives identification of the agent with the ground.

- (47) [VOICEP [the uninvited guests] [APPLP [~~the uninvited guests~~] [VP sandwiches stuffed]]]

The derived subject is the theme/figure argument of ordinary TVRs, but it is the ground argument if it is present, as in (46). On the theory proposed here, this follows from ordinary minimality restric-

tions on A-movement. The higher DP, the ground, will move to the specifier of Reflexive Voice if present; in its absence, the lower figure will instead. The movement trigger on Voice_{REFL} is indifferent as to the base position (and thus the base thematic role) of the lower element that will come to bind the free variable: whatever is closest moves, and what moves binds.

As a second point illustrating prediction (45b), consider Greek verbal reflexives formed with the reflexivizer *afto-*, which are obligatorily agent-oriented but not otherwise lexically restricted (Spathas et al., 2015; Embick, 2004b; Paparounas, 2023). When ditransitive predicates are reflexivized in Greek, the theme is identified with the agent across a goal; notably, the goal is never identified with the agent (see Alexiadou 2014). This fact correlates with an independent property of Greek goal DPs: they are A-inert, never being promoted under passivization (but acting as defective interveners in both passivization, Anagnostopoulou 2003b, and reflexivization, Paparounas 2023). If the analysis advanced here for Turkish were extended to Greek, this fact would receive an automatic explanation: because Greek dative goals never A-move, they can neither be promoted under passivization nor be identified with the agent in verbal reflexivization. Importantly, this explanation presupposes independent evidence for movement-based reflexivization in Greek, and is thus promissory.

Our account then makes available a way of understanding thematic variation in verbal reflexivization by tying it to an independent factor, namely, language-particular constraints on A-movement. This avenue is worth exploring; we consider it a sign of progress that it is eminently falsifiable.

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